SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105

Student Name: G. Venkata Niranjan Reddy **Reg. No.:**192525199

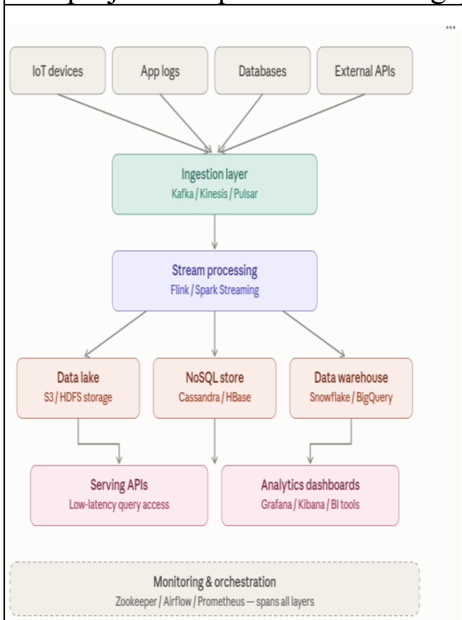
Course Code:CSA1522 **Slot:**C

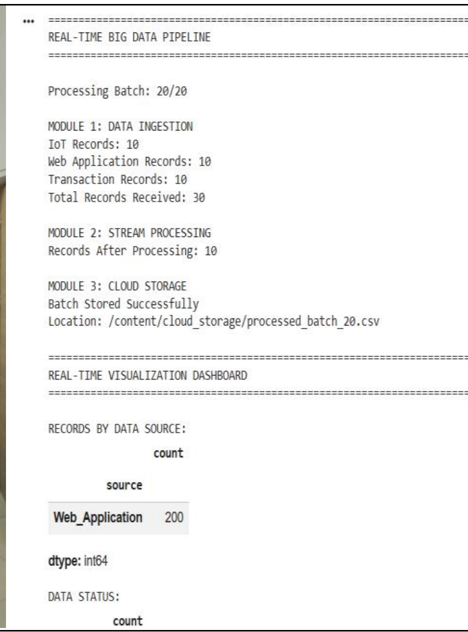
Course Name: Cloud Computing & Big Data Analytics

Course Faculty:DR.P.Rajaram

Project Title:

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)





Project Description: (here you write what you did in this project (contribution) including Model Description

Real-Time Big Data Ingestion Pipeline from Distributed Sources is a data engineering project designed to collect, process, store, and visualize large volumes of data generated from multiple distributed sources in real time. The system provides a unified pipeline for handling data from sources such as sensors, applications, log files, APIs, and databases.

The project continuously receives incoming data and processes it using a stream-processing mechanism to perform filtering, transformation, validation, and aggregation. The processed data is then stored in scalable cloud or distributed storage systems for further analysis. A visualization dashboard is used to display real-time information, trends, and important insights.

The proposed system consists of three major modules: **Multi-Source Data Ingestion**, **Stream Processing & Transformation**, and **Cloud Storage & Visualization**. The pipeline reduces manual data processing, improves data availability, supports real-time analytics, and provides a scalable solution for handling high-volume distributed data.

Student Signature **Guide Signature**